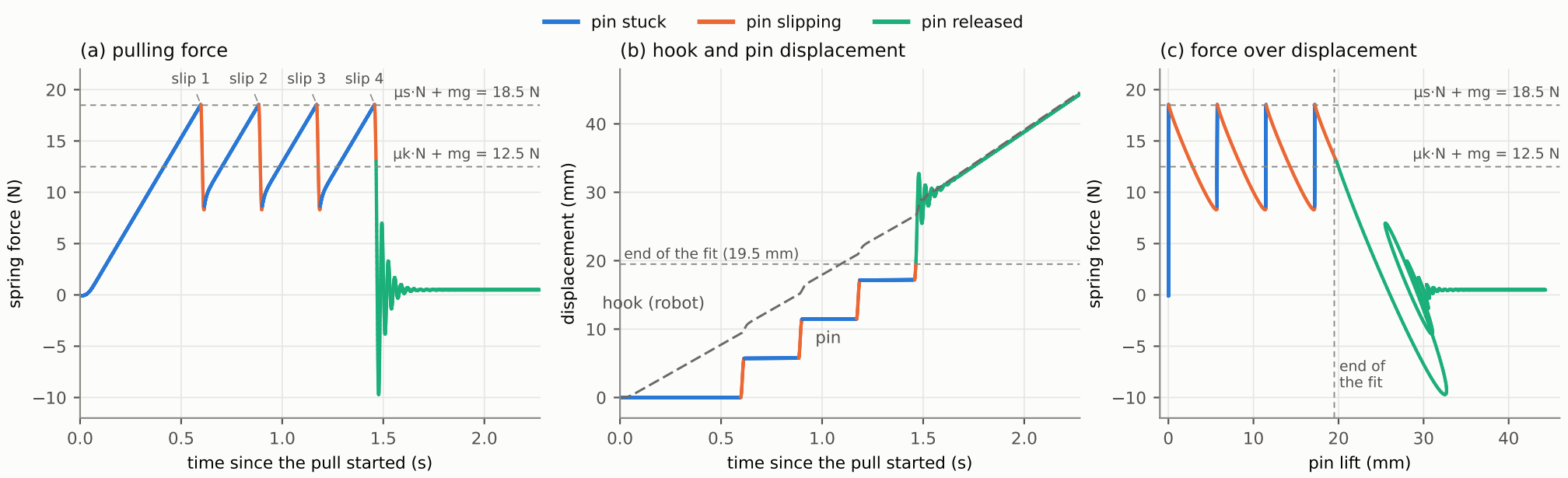

pin stuck
pin slipping
pin released
(a) pulling force
slip 1
slip 2
slip 3
slip 4
μs·N + mg = 18.5 N
μk·N + mg = 12.5 N
spring force (N)
time since the pull started (s)
(b) hook and pin displacement
end of the fit (19.5 mm)
hook (robot)
pin
displacement (mm)
time since the pull started (s)
(c) force over displacement
μs·N + mg = 18.5 N
μk·N + mg = 12.5 N
end of
the fit
spring force (N)
pin lift (mm)